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Decoding the DALY SmartBMS protocol

tanoshimini · Apr 27, 2021

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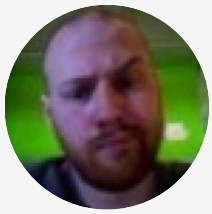


tanoshimini

Apr 27, 2021 #1

I have a some SmartBMS units from Daly that I want to talk to. On other threads in this forum, I've seen some discussion of the protocol, and there have been some successes that people have had in getting *some* documentation out of them. Having read over the

Solar Enthusiast
Joined: Mar 28, 2021
Messages: 110

documentation, I've come to discover that it only covers a few aspects of the BMS, and does not cover *at all,* anything to do with setting or retrieving it's configuration parameters.

To that end, I decided to use a Teensy 3.2 to monitor both sides of the communication between the BMS and the official Daly BMS software as it goes about setting and retrieving various parameters. Here's a raw dump of the back and forth I captured:

Normal chatter about cell voltages, etc. This stuff is covered in the documentation, and served as a sanity check.

Code:

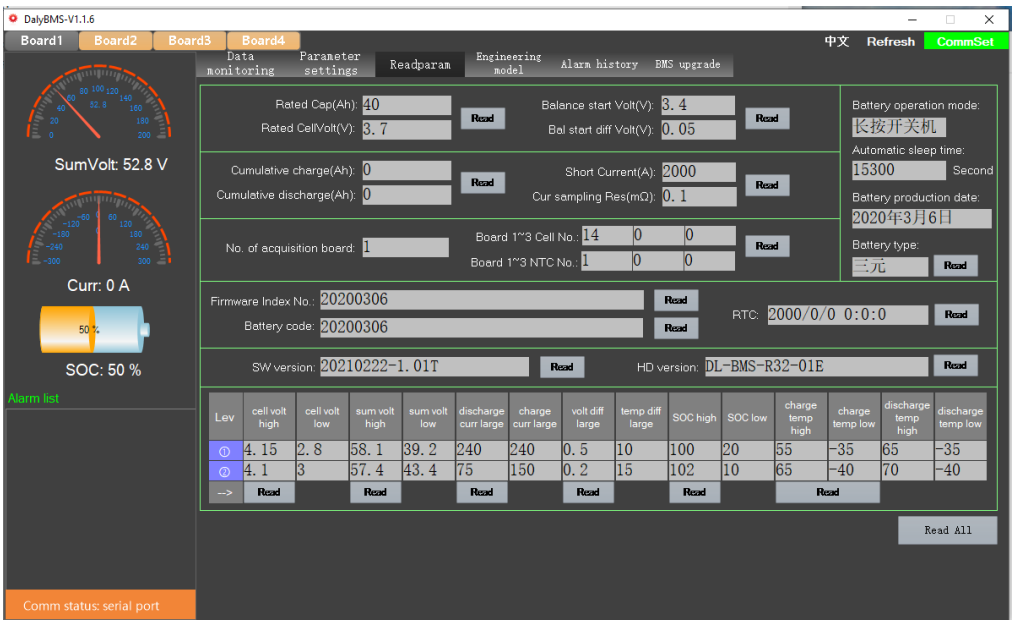
```
-> a5 40 90 08 00 00 00 00 00 00 00 00 7d
<- a5 01 90 08 02 10 00 00 75 30 01 f4 ea

-> a5 40 91 08 00 00 00 00 00 00 00 00 7e
<- a5 01 91 08 0e c8 05 0e 9c 09 01 f4 c2

-> a5 40 92 08 00 00 00 00 00 00 00 00 7f
<- a5 01 92 08 3d 01 3d 01 9c 09 01 f4 56

-> a5 40 93 08 00 00 00 00 00 00 00 00 80
<- a5 01 93 08 00 01 01 10 00 00 4e 20 c1

-> a5 40 94 08 00 00 00 00 00 00 00 00 81
<- a5 01 94 08 0e 01 00 00 02 00 00 20 73

-> a5 40 95 08 00 00 00 00 00 00 00 00 82
<- a5 01 95 08 01 0e c0 0e c2 0e c7 20 d7
<- a5 01 95 08 02 0e c8 0e c8 0e c6 20 e5
<- a5 01 95 08 03 0e c2 0e c2 0e 9c 20 b0
<- a5 01 95 08 04 0e c4 0e c4 0e c4 20 dd
<- a5 01 95 08 05 0e c3 0e c0 0e c4 20 d9

-> a5 40 96 08 00 00 00 00 00 00 00 00 83
<- a5 01 96 08 01 3d 00 00 00 00 00 00 82
<- a5 01 96 08 02 00 00 00 00 00 00 00 46

-> a5 40 97 08 00 00 00 00 00 00 00 00 84
<- a5 01 97 08 00 00 00 00 00 00 00 00 45

-> a5 40 98 08 00 00 00 00 00 00 00 00 85
<- a5 01 98 08 00 00 00 00 00 00 00 00 46

-> a5 40 d8 08 00 00 00 00 00 00 00 00 c5
<- a5 01 d8 08 00 00 00 00 00 00 00 00 86
```

This stuff is more interesting... It's the sequence that runs when you mash the "Read All" button on the "Read Settings" page of the official app. Attached below is a screenshot of the app after it's decoded this sequence. The app appears to helpfully group settings into things that can be retrieved with a single command, and these groups correspond 1-to-1 with the command sequence. It shouldn't be too hard to pick apart which one does what.

Code:

```
-> a5 40 50 08 00 00 00 00 00 00 00 00 3d // Query the rate
<- a5 01 50 08 00 00 9c 40 00 00 0e 74 5c

-> a5 40 5f 08 00 00 00 00 00 00 00 00 4c // Query the volt
<- a5 01 5f 08 0d 48 00 32 00 00 0e 74 16

-> a5 40 52 08 00 00 00 00 00 00 00 00 3f
<- a5 01 52 08 00 00 00 00 00 00 00 00 00

-> a5 40 60 08 00 00 00 00 00 00 00 00 4d // Query the shor
<- a5 01 60 08 07 d0 00 64 00 00 00 00 49

-> a5 40 51 08 00 00 00 00 00 00 00 00 3e // Query the "Num
<- a5 01 51 08 01 0e 00 00 01 00 00 00 0f

-> a5 40 53 08 00 00 00 00 00 00 00 00 40 // Query "Battery
<- a5 01 53 08 01 02 15 03 06 ff ff 00 20

-> a5 40 54 08 00 00 00 00 00 00 00 00 41 // ??? Query the
<- a5 01 54 08 32 30 32 30 30 33 30 36 8f

-> a5 40 57 08 00 00 00 00 00 00 00 00 44 // ??? Query the
<- a5 01 57 08 01 32 30 32 30 30 33 30 5d
<- a5 01 57 08 02 36 20 20 20 20 20 20 fd
<- a5 01 57 08 03 20 20 20 20 20 20 20 e8
<- a5 01 57 08 04 20 20 20 20 20 20 20 e9
<- a5 01 57 08 05 20 20 20 20 20 20 20 ea

-> a5 40 61 08 00 00 00 00 00 00 00 00 4e
<- a5 01 61 08 00 00 00 00 00 00 20 20 4f

-> a5 40 62 08 00 00 00 00 00 00 00 00 4f // Query Software
<- a5 01 62 08 01 32 30 32 31 30 32 32 6a
<- a5 01 62 08 02 32 2d 31 2e 30 31 54 85

-> a5 40 63 08 00 00 00 00 00 00 00 00 50 // Query Hardware
<- a5 01 63 08 01 44 4c 2d 42 4d 53 2d de
```

```
<- a5 01 63 08 02 52 33 32 2d 30 31 45 9d

-> a5 40 59 08 00 00 00 00 00 00 00 00 46 // Query the Leve
<- a5 01 59 08 10 36 10 04 0a f0 0b b8 1e

-> a5 40 5a 08 00 00 00 00 00 00 00 00 47 // Query the Leve
<- a5 01 5a 08 02 45 02 3e 01 88 01 b2 cb

-> a5 40 5b 08 00 00 00 00 00 00 00 00 48 // Query the Leve
<- a5 01 5b 08 6b d0 6f 54 7e 90 78 1e ab

-> a5 40 5e 08 00 00 00 00 00 00 00 00 4b // Query the Leve
<- a5 01 5e 08 01 f4 00 c8 0a 0f 78 1e 78

-> a5 40 5d 08 00 00 00 00 00 00 00 00 4a
<- a5 01 5d 08 03 e8 03 fc 00 c8 00 64 21

-> a5 40 5c 08 00 00 00 00 00 00 00 00 49
<- a5 01 5c 08 5f 69 05 00 69 6e 05 00 b3
```



Likewise, this is a dump of the sequence that from mashing the "Set All" button on the "Parameter Settings" page. It *looks* like they largely mirror the commands being send in the *read* section, and it shouldn't be too much work to prove that out.

Code:

```
-> a5 40 10 08 00 00 9c 40 00 00 0e 74 5b // Set the rated
<- a5 01 10 08 01 10 00 00 75 30 01 f4 69

-> a5 40 1f 08 0d 48 00 32 00 00 0e 74 15 // Set the voltag
```

```
<- a5 01 1f 08 01 10 00 00 75 30 01 f4 78

-> a5 40 12 08 00 00 00 00 00 00 00 00 ff
<- a5 01 12 08 01 10 00 00 75 30 01 f4 6b

-> a5 40 20 08 07 d0 00 64 00 00 00 00 48 // Set the short-
<- a5 01 20 08 01 10 00 00 75 30 01 f4 79

-> a5 40 11 08 01 0e 00 00 01 00 00 00 0e // Set the "Numbe
<- a5 01 11 08 01 10 00 00 75 30 01 f4 6a

-> a5 40 13 08 01 02 15 03 06 ff ff 00 1f // Set "Battery c
<- a5 01 13 08 01 10 00 00 75 30 01 f4 6c

-> a5 40 14 08 32 30 32 30 30 33 30 36 8e // ??? Set the "F
<- a5 01 14 08 01 10 00 00 75 30 01 f4 6d

-> a5 40 17 08 01 32 30 32 30 30 33 30 5c // ??? Set the "E
-> a5 40 17 08 02 36 20 20 20 20 20 20 fc
-> a5 40 17 08 03 20 20 20 20 20 20 20 e7
-> a5 40 17 08 04 20 20 20 20 20 20 20 e8
-> a5 40 17 08 05 20 20 20 20 20 20 20 e9
<- a5 01 17 08 01 10 00 00 75 30 01 f4 70

-> a5 40 19 08 10 36 10 04 0a f0 0b b8 1d // Set the Level
<- a5 01 19 08 01 10 00 00 75 30 01 f4 72

-> a5 40 1a 08 02 44 02 3e 01 88 01 b2 c9 // Set the Level
<- a5 01 1a 08 01 10 00 00 75 30 01 f4 73

-> a5 40 1b 08 6b d0 6f 54 7e 90 78 1e aa // Set the Level
<- a5 01 1b 08 01 10 00 00 75 30 01 f4 74

-> a5 40 1e 08 01 f4 00 c8 0a 0f 78 1e 77 // Set the Level
<- a5 01 1e 08 01 10 00 00 75 30 01 f4 77

-> a5 40 1d 08 03 e8 03 fc 00 c8 00 64 20
<- a5 01 1d 08 01 10 00 00 75 30 01 f4 76

-> a5 40 1c 08 5f 69 05 00 69 6e 05 00 b2
<- a5 01 1c 08 01 10 00 00 75 30 01 f4 75
```

Last edited: Apr 27, 2021

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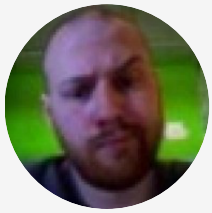
curiouscarbon and Just John



tanoshimini
Solar Enthusiast
Joined: Mar 28, 2021
Messages: 110

Apr 27, 2021 #2

Command **0x62** appears to query the "Software Version":

Code:

```
-> a5 40 62 08 00 00 00 00 00 00 00 00 4f
<- a5 01 62 08 01 32 30 32 31 30 32 32 6a // "2021022"
<- a5 01 62 08 02 32 2d 31 2e 30 31 54 85 // "2-1.01T"
```

- **Byte 0**: The frame number; Each frame carries 7 bytes of the string.
- **Byte 1-7:** The characters that make up the version string.

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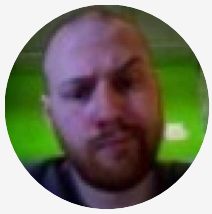
curiouscarbon and Just John



tanoshimini
Solar Enthusiast
Joined: Mar 28, 2021
Messages: 110

Apr 27, 2021 #3

Command **0x63** appears to query the "Hardware Version"

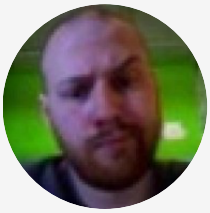
Code:

```
-> a5 40 63 08 00 00 00 00 00 00 00 00 50
<- a5 01 63 08 01 44 4c 2d 42 4d 53 2d de // "DL-BMS-"
<- a5 01 63 08 02 52 33 32 2d 30 31 45 9d // "R32-01E"
```

- **Byte 0**: The frame number; Each frame carries 7 bytes of the string.
- **Byte 1-7:** The characters that make up the version string.

Report Like Quote Reply

curiouscarbon and Just John



tanoshimini
Solar Enthusiast
Joined: Mar 28, 2021
Messages: 110

Apr 27, 2021 #4

Command **0x51** appears to query the "Number of acquisition board", "Cell counts" and "Temp Sensor counts"

Code:

```
-> a5 40 51 08 00 00 00 00 00 00 00 00 3e
<- a5 01 51 08 01 0e 00 00 01 00 00 00 0f
```

- **Byte 0:** Number of acquisition board? [01] = Board count of 1
- **Byte 1-3:** Cell counts [0E 00 00] = (14, 0, 0)
- **Byte 4-6:** Temp sensor (NTC thermistor) counts [01 00 00] = (1, 0, 0)
- **Byte 7**: Reserved?

Last edited: Apr 27, 2021

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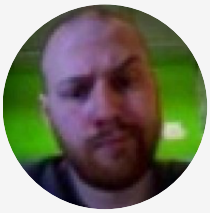
Robertp23, curiouscarbon and Just John



tanoshimini
Solar Enthusiast
Joined: Mar 28, 2021
Messages: 110

Apr 27, 2021 #5

Command **0x59** appears to query the Level 1 and 2 alarm thresholds for high and low cell voltages.

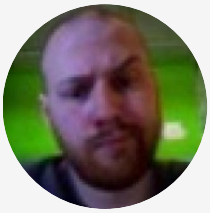
Code:

```
-> a5 40 59 08 00 00 00 00 00 00 00 00 46
<- a5 01 59 08 10 36 10 04 0a f0 0b b8 1e
```

- **Byte 0-1:** Level-1 alarm threshold for Cell High Voltage, in millivolts [10 36] = 4.15v
- **Byte 2-3:** Level-2 alarm threshold for Cell High Voltage, in millivolts [10 04] = 4.1v
- **Byte 4-5**: Level-1 alarm threshold for Cell Low Voltage, in millivolts [0A F0] = 2.8v
- **Byte 6-7**: Level-2 alarm threshold for Cell Low Voltage, in millivolts [0B B8] = 3.0v

Report Like Quote Reply

curiouscarbon and Just John



tanoshimini
Solar Enthusiast
Joined: Mar 28, 2021
Messages: 110

Apr 27, 2021 #6

Command **0x5A** appears to query the Level 1 and 2 alarm thresholds for high and low voltages for the pack as a whole.

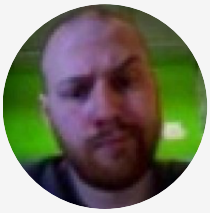
Code:

```
-> a5 40 5a 08 00 00 00 00 00 00 00 00 47
<- a5 01 5a 08 02 45 02 3e 01 88 01 b2 cb
```

- **Byte 0-1:** Level-1 alarm threshold for Pack High Voltage, in 10th's of a volt [02 45] = 58.1v
- **Byte 2-3:** Level-2 alarm threshold for Pack High Voltage, in 10th's of a volt [02 3E] = 57.4v
- **Byte 4-5**: Level-1 alarm threshold for Pack Low Voltage, in 10th's of a volt [01 88] = 39.2v
- **Byte 6-7**: Level-2 alarm threshold for Pack Low Voltage, in 10th's of a volt [01 B2] = 43.4v

Report Like Quote Reply

curiouscarbon and Just John



tanoshimini
Solar Enthusiast
Joined: Mar 28, 2021
Messages: 110

Apr 27, 2021 #7

Command **0x5E** appears to query the Level 1 and 2 alarm thresholds for allowable difference in cell voltage and temperature sensor readings

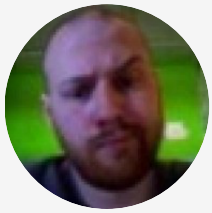
Code:

```
-> a5 40 5e 08 00 00 00 00 00 00 00 00 4b
<- a5 01 5e 08 01 f4 00 c8 0a 0f 78 1e 78
```

- **Byte 0-1:** Level-1 alarm threshold for Cell Voltage Difference, in millivolts [01 F4] = 0.5v
- **Byte 2-3:** Level-2 alarm threshold for Cell Voltage Difference, in millivolts [00 C8] = 0.2v
- **Byte 4**: Level-1 alarm threshold for Temp Sensor Difference, in °C [0A] = 10°C
- **Byte 5**: Level-2 alarm threshold for Temp Sensor Difference, in °C [0F] = 15°C
- **Byte 6-7:** Reserved?

Report Like Quote Reply

curiouscarbon and Just John



tanoshimini
Solar Enthusiast
Joined: Mar 28, 2021
Messages: 110

Apr 27, 2021 #8

Command **0x5B** appears to query the Level 1 and 2 alarm thresholds for charge and discharge current for the pack.

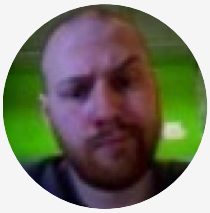
Code:

```
-> a5 40 5b 08 00 00 00 00 00 00 00 00 48
<- a5 01 5b 08 6b d0 6f 54 7e 90 78 1e ab
```

- **Byte 0-1:** Level-1 alarm threshold for High Charge Current, in 10ths of an amp (with a zero-value of 30000) [6B D0] = 240A; (30000 - (240 * 10))
- **Byte 2-3:** Level-2 alarm threshold for High Charge Current, in 10ths of an amp (with a zero-value of 30000) [6F 54] = 150A; (30000 - (150 * 10))
- **Byte 4-5**: Level-1 alarm threshold for High Discharge Current, in 10ths of an amp (with a zero-value of 30000) [7E 90] = 240A; (30000 + (240 * 10))
- **Byte 6-7**: Level-2 alarm threshold for High Discharge Current, in 10ths of an amp (with a zero-value of 30000) [78 1e] = 75A; (30000 + (75* 10))

Report Like Quote Reply

curiouscarbon and Just John



tanoshimini
Solar Enthusiast
Joined: Mar 28, 2021
Messages: 110

Apr 27, 2021 #9

Command **0x50** appears to query the rated pack capacity and nominal cell voltage.

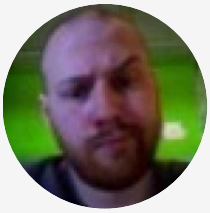
Code:

```
-> a5 40 50 08 00 00 00 00 00 00 00 00 3d
<- a5 01 50 08 00 00 9c 40 00 00 0e 74 5c
```

- **Byte 0-3:** Rated pack capacity, in mAh. [00 00 9C 40] = 40Ah
- **Byte 4-5**: Reserved?
- **Byte 6-7**: Nominal cell voltage, in millivolts. [0E 74] = 3.7v

Report Like Quote Reply

Robertp23, curiouscarbon and Just John



tanoshimini
Solar Enthusiast
Joined: Mar 28, 2021
Messages: 110

Apr 27, 2021 #10

Command **0x5F** appears to query the voltage thresholds that control balancing.

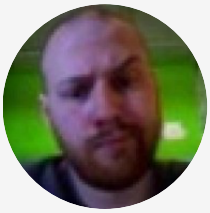
Code:

```
-> a5 40 5f 08 00 00 00 00 00 00 00 00 4c
<- a5 01 5f 08 0d 48 00 32 00 00 0e 74 16
```

- **Byte 0-1**: Cell voltage threshold at which balancing is enabled, in millivolts. [0D 48] = 3.4v
- **Byte 2-3**: Voltage differential between cells that is considered acceptable, in millivolts. [00 32] = 0.05v (The software doesn't seem to allow a value *below* 0.05v, but the protocol would seem to allow it. *Interesting*.)
- **Byte 4-7**: Reserved?

Report Like Quote Reply

curiouscarbon and Just John



tanoshimini
Solar Enthusiast
Joined: Mar 28, 2021
Messages: 110

Apr 27, 2021 #11

Command **0x60** appears to query the short-circuit shutdown threshold and the current sampling resolution.

Code:

```
-> a5 40 60 08 00 00 00 00 00 00 00 00 4d
<- a5 01 60 08 07 d0 00 64 00 00 00 00 49
```

- **Byte 0-1**: Short-circuit shutdown threshold, in amps. [07 D0] = 2000A
- **Byte 2-3**: Current sampling resistance, in milliohms. [00 64] = 0.1Ω
- **Byte 4-7**: Reserved?

Report Like Quote Reply

curiouscarbon and Just John

J

Just John
Solar Wizard
Joined: Aug 15, 2020
Messages: 2,836
Location: AZ

Apr 27, 2021 #12

Excellent, I haven't had time to set up to sniff/capture the traffic myself. I suspect Seiko might have a data sheet somewhere, since they advertise it as using a Seiko chip. This is great information, thank you for sharing.

Now to figure out if my 12v and 24v share the same command/response set, likely they do.

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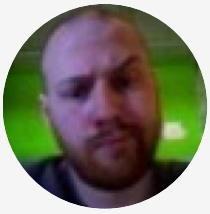


tanoshimini
Solar Enthusiast
Joined: Mar 28, 2021
Messages: 110

Apr 28, 2021 #13

Does anyone know what seiko chip they're using? I can't seem to find much information on that. Knowing the chip, there may be manufacturer datasheets and documentation that I can mine for additional features or functionality that Daly chose not to expose.

Report Like Quote Reply

Apr 28, 2021 #14

Just John
Solar Wizard
Joined: Aug 15, 2020
Messages: 2,836
Location: AZ

> Does anyone know what seiko chip they're using? I can't seem to find much information on that. Knowing the chip, there may be manufacturer datasheets and documentation that I can mine for additional features or functionality that Daly chose not to expose.

I would like to know as well, I haven't found anything published by Daly that tells us the specific chip, unfortunately.

Report Like Quote Reply

TomC4306

Apr 29, 2021 #15

jblance
New Member
Joined: Apr 14, 2020
Messages: 3

This might of use https://github.com/jblance/mpp-solar/blob/master/docs/protocols/DALY-Daly_RS485_UART_Protocol.pdf

Report Like Quote Reply

Just John

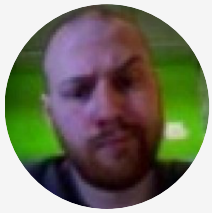
Apr 29, 2021 #16

Just John
Solar Wizard
Joined: Aug 15, 2020
Messages: 2,836
Location: AZ

> jblance said:
>
> This might of use https://github.com/jblance/mpp-solar/blob/master/docs/protocols/DALY-Daly_RS485_UART_Protocol.pdf

Looks like useful information despite the "translation".

Report Like Quote Reply



tanoshimini
Solar Enthusiast
Joined: Mar 28, 2021
Messages: 110

Apr 29, 2021 #17

> jblance said:
>
> This might of use https://github.com/jblance/mpp-solar/blob/master/docs/protocols/DALY-Daly_RS485_UART_Protocol.pdf

Good find! This seems to be another version of the documentation that I had found already, and serves to confirm and clarify a few things.

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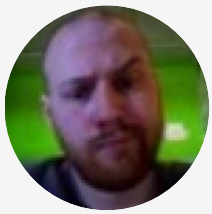
curiouscarbon
Science Penguin
Joined: Jun 30, 2020
Messages: 3,046

Apr 29, 2021 #18

this thread is legendary. amazing work @jallum

Report Like Quote Reply

OffGridGuy and tanoshimini



tanoshimini
Solar Enthusiast
Joined: Mar 28, 2021
Messages: 110

May 2, 2021 #19

I took some inspiration from the SimpBMS for tesla modules, and I now have the Daly sending BMS.Can messages that my Victron equipment is reading successfully. Yay! From what I understand this would make my little adapter work with any inverter that is compatible with the SimpBMS or the ones from REC.

I'm going to streamline the code and push it to GitHub in the next day or so.

Report Like Quote Reply

binair, Just John and curiouscarbon

BlueFox
New Member
Joined: Mar 12, 2021
Messages: 41

May 2, 2021 #20

I wish all these chinese BMS brands would just use standard protocols like Modbus or SMbus, would make things so much easier!

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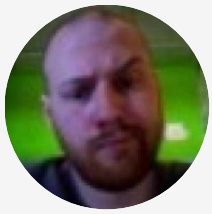

tanoshimini
Solar Enthusiast
Joined: Mar 28, 2021
Messages: 110

May 2, 2021 #21

> BlueFox said:
>
> I wish all these chinese BMS brands would just use standard protocols like Modbus or SMbus, would make things so much easier!

Naw... we're still firmly in the BETA/VHS (or Blue-Ray/HD-DVD for younger people) stage of this tech. It's okay, though, give it a little while and it'll firm up. In the meantime, these little microcontrollers make for a nifty fix to the problem.

I'm working through the various messages and seeing what kind of junk I'm sending the Victron (Cerbo GX) through it's UI / logging. As I find things that are missing or wrong, I'm rigging those messages 1st, with dummy / hard-coded values — just to see them come through and to work out the encoding / formatting of the message. 2nd, I'm wiring those messages up to real values as read from the Daly. 3rd, I'm cleaning up the code and refactoring it before I push it up to Github.

I'm going to try and build the microcontroller software in such a way that it could be easily adapted to pull from other types of BMS, run on other flavors of micro controller and perform the same general services. I've looked around, and all the things I've found (so far) are so hopelessly hard-coded with spaghetti logic. I'm pretty certain that it can be done better.

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PowerCell, lexio, curiouscarbon and 1 other person

Just John
Solar Wizard
Joined: Aug 15, 2020
Messages: 2,836
Location: AZ

May 3, 2021 #22

> jallum said:
>
> I took some inspiration from the SimpBMS for tesla modules, and I now have the Daly sending BMS.Can messages that my Victron equipment is reading successfully. Yay! From what I understand this would make my little adapter work with any inverter that is compatible with the SimpBMS or the ones from REC.
>
> I'm going to streamline the code and push it to GitHub in the next day or so.

Nice! I am very interested in this project, getting a Daly to talk to anything is an achievement, a Victron is fantastic.

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parv
New Member
Joined: Nov 30, 2020
Messages: 1

Aug 11, 2021 #23

> tanoshimini said:
>
> I have a some SmartBMS units from Daly that I want to talk to. On other threads in this forum, I've seen some discussion of the protocol, and there have been some successes that people have had in getting *some* documentation out of them. Having read over the documentation, I've come to discover that it only covers a few aspects of the BMS, and does not cover
>
> Click to expand...

Need help in decoding battery code!! Thanks in advance

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birdie1234
New Member
Joined: Jun 20, 2021
Messages: 2

Aug 24, 2021 #24

I just startet reverse engineering the daly/sinowealth protocol and took the 3 byte protocol from the sinowealth bms tool.

If you are interested (Will add more parameter the next days):
https://github.com/birdie1/bms-reader

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jmjordan
New Member
Joined: Aug 25, 2021
Messages: 1

Aug 25, 2021 #25

I found this Python based tool that works with both regular Daly and Sinowealth versions. It worked out of the box for me using the UART to serial cable. It also supports RS485.

GitHub - dreadnought/python-daly-bms: Python modul…
Python module for Daly BMS devices. Contribute to dreadnought/python-daly-bms development by creating an account on GitHub.
github.com

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elfrisky

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